Physics IA Experiment 2003

Topic: To make measurements of a ball rolling down a slope in order to calculate its acceleration.

Independent variable: different distances, s
Dependent variable : time taken for the ball to roll down

variable	Actual uncertainty	Percentage uncertainty/%
Metre rule	$\pm$ 0.01	1%
Half-metre rule	$\pm$ 0.01	2%
stopwatch	$\pm$ 0.01s	1%

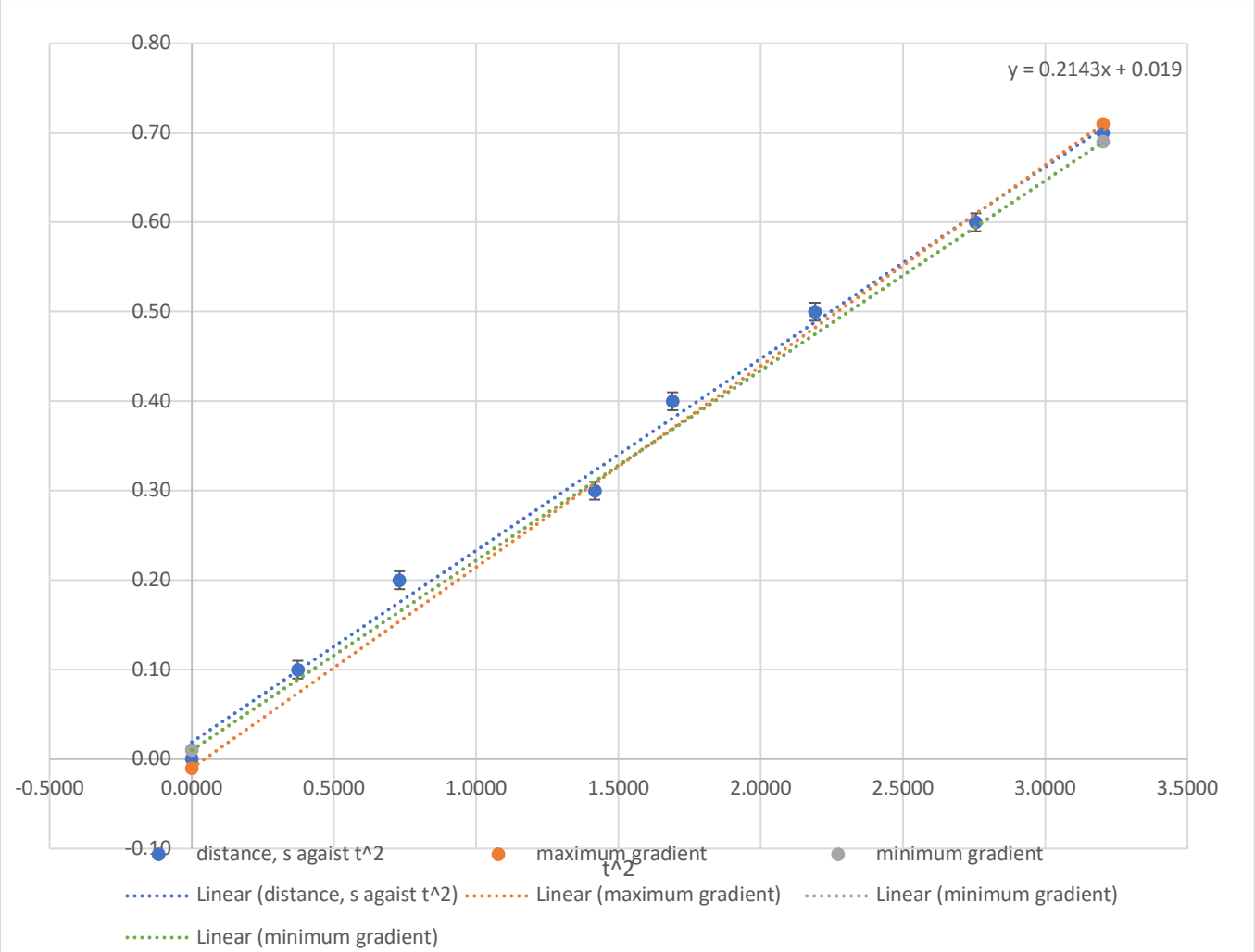
Readings

Distance, s / m $\pm$	t1 / s $\pm$ 0.01	t2/ s $\pm$ 0.01	t3/ s $\pm$ 0.01	t avg / s $\pm$ 0.01	$t^2/\ s^2$ $\pm$ 0.0001
0.00	0.00	0.00	0.00	0.00	0.0000
0.10	0.53	0.60	0.69	0.61	0.3721
0.20	0.90	0.88	0.81	0.86	0.7396
0.30	1.22	1.22	1.13	1.19	1.4161
0.40	1.28	1.28	1.35	1.30	1.6900
0.50	1.53	1.44	1.47	1.48	2.1904
0.60	1.56	1.72	1.69	1.66	2.7556
0.70	1.82	1.78	1.78	1.79	3.2041

Distance,s /m	0.00	0.10	0.20	0.30	0.40	0.50	0.60	0.70
t^2/s^2	0.0000	0.3721	0.7296	1.4161	1.6900	2.1904	2.7556	3.2041

s = ut + $\frac{1}{2}at^2$
Since u = 0 ,
s = $\frac{1}{2}at^2$
Gradient = ½ a
= 0.2143
Value of a from graph = 0.4286

Height of the slope = 7 cm



Data Processing	Error Propogation
When s= 0.00, $t^2 = 0.0000$ S = ½at^2 0.0 = 1/2a$(0.000)^2$ a = 0	$\%\frac{\Delta a}{a} = (\frac{\Delta t^2}{t^2} + \frac{\Delta s}{s}) \times 100\%$ $=(\frac{0.0001}{0.0000} + \frac{0.01}{0.00}) \times 100\%$ $= 0.00\% \approx 0.00\%$
When s= 0.10, , $t^2 = 0.3721$ s =1/2at^2 a = $\frac{2(0.10)}{0.3721}$ = 0.5374 ≈ 0.54	$\%\frac{\Delta a}{a} = (\frac{\Delta t^2}{t^2} + \frac{\Delta s}{s}) \times 100\%$ $=(\frac{0.0001}{0.3721} + \frac{0.01}{0.10}) \times 100\%$ $= 10.026\% \approx 10.0\%$

When s= 0.20, , $t^2 = 0.3721$ s =1/2at^2 a = $\frac{2(0.20)}{0.7396}$ = 0.54083 ≈ 0.54	$\%\frac{\Delta a}{a} = (\frac{\Delta t^2}{t^2} + \frac{\Delta s}{s}) \times 100\%$ =($\frac{0.0001}{0.7396} + \frac{0.01}{0.20}$)× 100% = 6.352% ≈ 6.35%
When s= 0.30, , $t^2 = 1.4161$ s =1/2at^2 a = $\frac{2(0.30)}{1.4161}$ = 0.4236 ≈ 0.42	$\%\frac{\Delta a}{a} = (\frac{\Delta t^2}{t^2} + \frac{\Delta s}{s}) \times 100\%$ =($\frac{0.0001}{0.7396} + \frac{0.01}{0.30}$)× 100% = 3.3468% ≈ 3.35%
When s= 0.40, , $t^2 = 1.6900$ s =1/2at^2 a = $\frac{2(0.40)}{1.6900}$ = 0.4733 ≈ 0.47	$\%\frac{\Delta a}{a} = (\frac{\Delta t^2}{t^2} + \frac{\Delta s}{s}) \times 100\%$ =($\frac{0.0001}{1.6900} + \frac{0.01}{0.40}$)× 100% = 2.5059 % ≈ 2.51%
When s= 0.50, , $t^2 = 2.1904$ s =1/2at^2 a = $\frac{2(0.50)}{2.1904}$ = 0.4565 ≈ 0.46	$\%\frac{\Delta a}{a} = (\frac{\Delta t^2}{t^2} + \frac{\Delta s}{s}) \times 100\%$ =($\frac{0.0001}{2.1904} + \frac{0.01}{0.50}$)× 100% =2.0045 % ≈ 2.00%
When s= 0.60, , $t^2 = 2.7556$ s =1/2at^2 a = $\frac{2(0.60)}{2.7556}$ = 0.4354 ≈ 0.44	$\%\frac{\Delta a}{a} = (\frac{\Delta t^2}{t^2} + \frac{\Delta s}{s}) \times 100\%$ =($\frac{0.0001}{2.7556} + \frac{0.01}{0.60}$)× 100% = 1.6702 % ≈ 1.67 %
When s= 0.70, , $t^2 = 3.2041$ s =1/2at^2 a = $\frac{2(0.70)}{3.2041}$ = 0.4369 ≈ 0.44	$\%\frac{\Delta a}{a} = (\frac{\Delta t^2}{t^2} + \frac{\Delta s}{s}) \times 100\%$ =($\frac{0.0001}{3.2041} + \frac{0.01}{0.70}$)× 100% = 1.4316 % ≈ 1.43%

a= $\frac{0+0.54+0.54+0.42+0.47+0.46+0.44+0.44}{8}$ = 0.41299 ≈ 0.41m/s^2

percentage uncertainty = $\frac{0+10.0+6.35+3.35+2.51+2.00+1.67+1.43}{8}$ = 3.4171 ≈ 3.41%

value of a from data processing = 0.41± 0.01

Expected acceleration of the tennis ball = gsinθ

= 9.81 × $\frac{7}{75}$

= 0.9156

≈ 0.92

Value of acceleration of the tennis ball from experiment is different from the value calculated due to errors such as systematic errors. Air resistance is not negligible. Thus net force is smaller resulting in acceleration to be smaller.

Errors

- Basket used to elevate the rulers to make plank has uneven surface affected the reading for height of slope.
- Rulers are quite thick so affects the reading for height of slope
- Human reaction time so time taken to roll down is shorter
- Tennis ball have uneven surface causing it to take longer time to roll down.
- Ball rolls to the end of ruler causing zero error
- Tennis ball does not roll immediately causing a time lag so time taken is shorter.
- Ruler has uneven surface preventing the ball from rolling down.

Measurement of time taken is the largest source of error due to reaction time, time lag. The time taken is longer due to the above. Causing a large percentage of error which affects the accuracy of the result.

Suggestions to minimise errors

- Use a basket or something big with a smoother surface to get a more accurate height reading.
- Use a ruler which is thinner yet strong to get a more accurate reading of height.
- Prevent the ball from rolling down to the end of ruler by placing a stopper at zero marking.
- Repeat the experiment two more times to reduce to error from reaction time.
- Use a tennis ball with a smoother surface to get a accurate reading of time taken